

MATH298B: DATA IN STEM
RESEARCH REPORT

Analysis of CO₂ Levels Over Time

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INTRODUCTION

Since 1958, the Mauna Loa Observatory in Hawaii has provided the longest continuous record of atmospheric CO₂ measurements, serving as a critical benchmark for understanding climate change. In this project, we analyze over five decades of Mauna Loa CO₂ data to answer a fundamental question: Is atmospheric CO₂ growing linearly or quadratically?

Distinguishing between these growth patterns carries profound implications for climate science and global policy. Linear growth indicates that CO₂ levels are increasing at a constant rate: the same number of parts per million (ppm) each year (stable and predictable trajectory). Quadratic growth reveals an accelerating trend where the rate of increase itself is growing over time, implying that CO₂ production is intensifying rather than stabilizing.

A quadratic growth model is far more alarming. Without intervention, atmospheric CO₂ will reach critical thresholds much faster under accelerating growth. The Intergovernmental Panel on Climate Change (IPCC) states that the "safe-ish" upper bound is about 1.5°C and classifies 2.0°C as the *systemic, irreversible, globally cascading "oh sh*t"-bound*. To stay below those thresholds, climate scientists have established carbon budgets (finite pools of CO₂ we can emit before exceeding these targets). Under quadratic growth, those budgets would shrink tremendously, compressing the timeline for climate action. Misidentifying the growth pattern of CO₂ could lead to underestimating the urgency of climate action, making this analysis a critical component of evidence-based climate policy.

DATA OVERVIEW

The dataset from the Mauna Loa Observatory contains monthly atmospheric CO₂ measurements spanning from 1960 to 2019. This location provides one of the most reliable long-term records of atmospheric carbon dioxide due to its remote volcanic summit, which minimizes local pollution interference. The data consists of 2,152 observations with two variables: Year (decimal format) and CO₂ concentration (ppm), which is graphically represented in Figure 1.

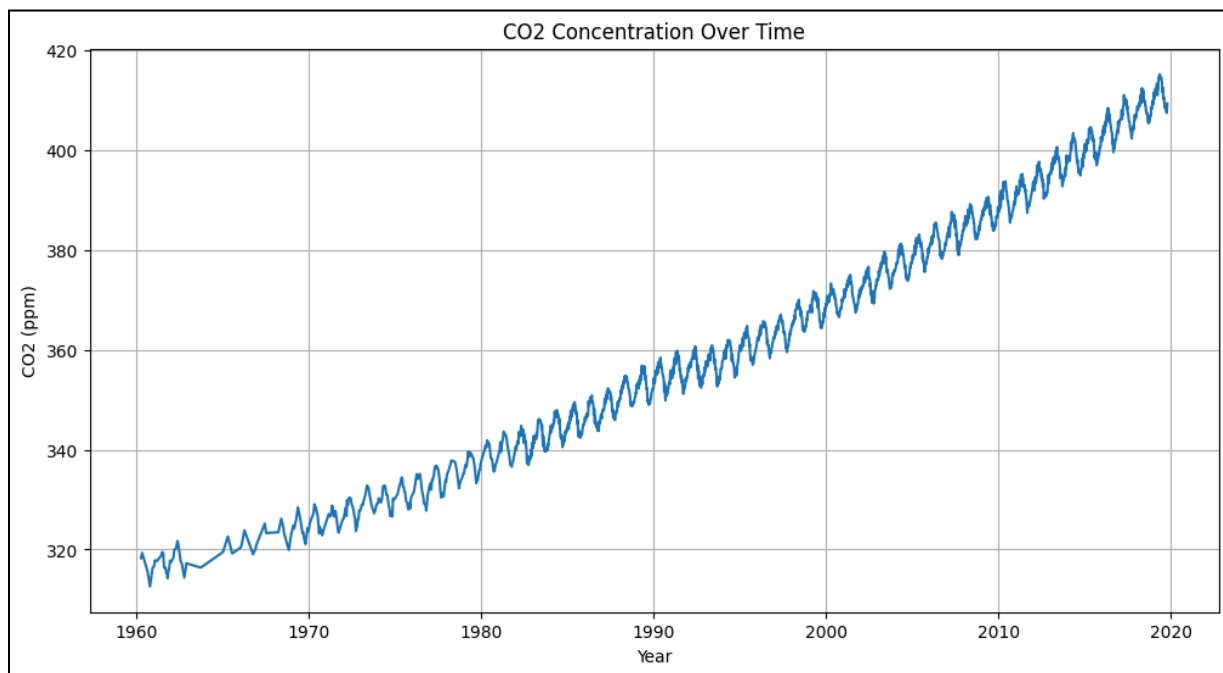


Figure 1: Graph of CO₂ concentration over time (1960-2019)

The CO₂ data reveal an interesting pattern of annual oscillations overlaid on a long-term upward trend. These oscillations can be explained by the seasonal growth and decay cycles of vegetation in the Northern Hemisphere, which contains the majority of the Earth's land mass and biomass. During spring and summer, plants enter their growing season and actively absorb CO₂ through photosynthesis, causing atmospheric CO₂ concentrations to decrease. During fall and winter, plants enter dormancy, decompose, and release stored CO₂ back into the atmosphere. This creates the “sawtooth” pattern visible in the data, with an amplitude of approximately 5-7 ppm.

METHODOLOGY

These seasonal swings are large relative to the year-to-year variations, obscuring the underlying growth trend. To properly fit a trend line for the long-term CO₂ change, we needed to remove these oscillations. We used a Fast Fourier Transform (FFT) approach: we computed the FFT of the entire CO₂ time series to identify dominant frequencies in the power spectrum. The power spectrum plot below (Figure 2) reveals a dominant frequency spike at approximately 1 cycle per year, corresponding to the annual vegetation cycle we identified earlier.

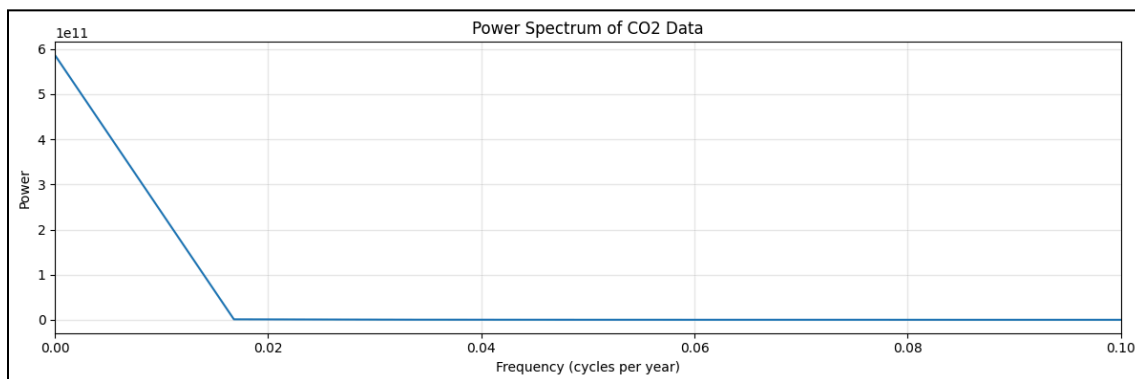


Figure 2: Power spectrum of CO₂ Data

Once we identified this dominant frequency, we selectively zeroed out the frequencies corresponding to the seasonal cycle in the FFT domain. We then applied the inverse FFT to reconstruct the signal without these oscillatory components. This frequency-filtering approach preserves the overall shape of the data while cleanly removing the periodic noise, allowing us to analyze the actual growth trend without seasonal variance interference. The resulting detrended CO₂ signal is displayed in Figure 3.

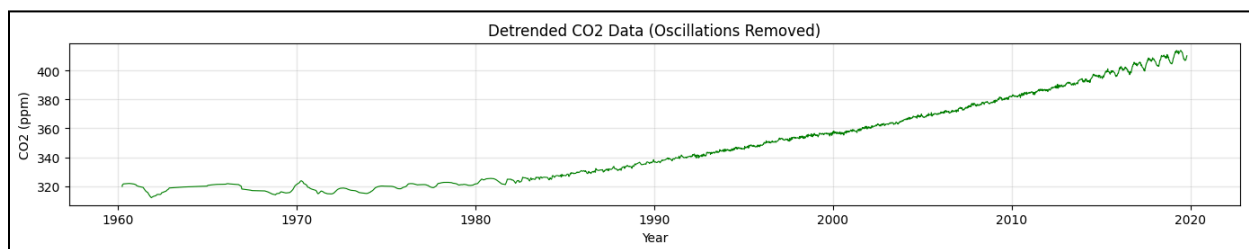


Figure 3: Detrended CO₂ Data

To examine the underlying trend in atmospheric CO₂ concentration, we performed least squares regression on the detrended data to test the two competing models: a linear model ($\text{CO}_2 = a + b * \text{Year}$) and a quadratic model ($\text{CO}_2 = a + b * \text{Year} + c * \text{Year}^2$). To determine which model best describes the data, we compared their “goodness-of-fit” using R^2 and Sum of Squared Residuals tests. Figure 4 displays both fitted models overlaid on the detrended CO₂ data, while Figure 5 plots both models' residuals. Table 1 is a summary of the results for the two testing methods.

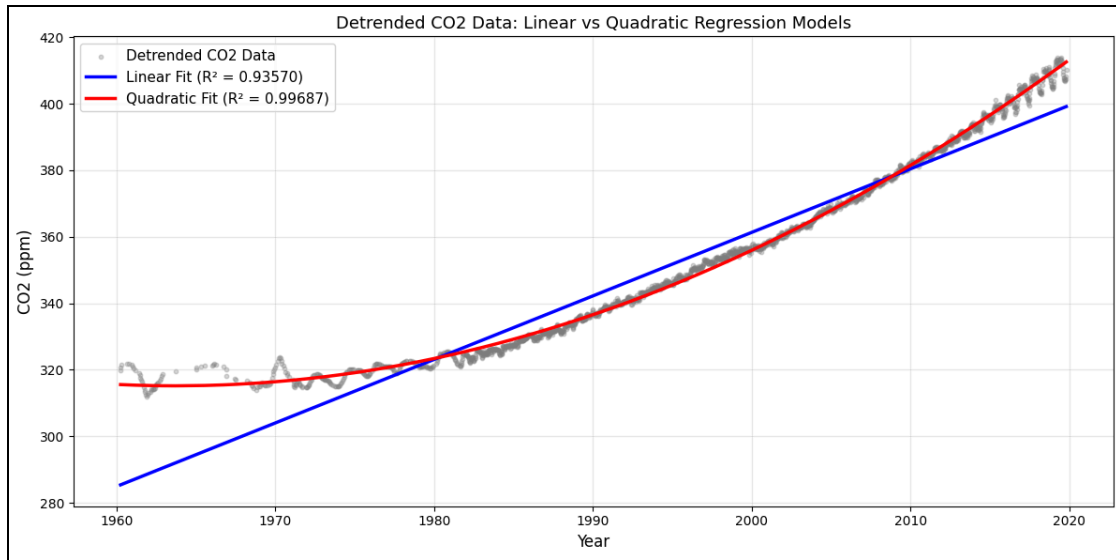


Figure 4: Fitted Models on Detrended Data (R^2 value in top left)

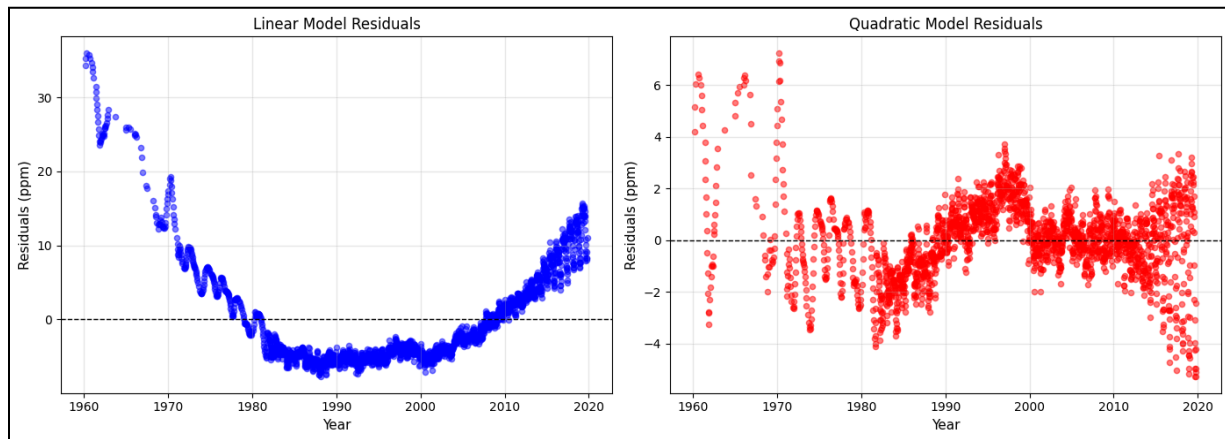


Figure 5: Linear and Quadratic Residuals

Metric	Linear Model	Quadratic Model
Equation	$(CO_2 = a + b * Year)$	$(CO_2 = a + b * Year + c * Year^2)$
R^2	0.9357	0.9969
Sum of Squared Residuals	106,189.14	5,173.64

Table 1: Results Summary

CONCLUSION

The quadratic model provides a substantially better fit to the data. With an R^2 value of 0.9969 compared to 0.9357, and residuals that are roughly one-twentieth the size, the evidence clearly indicated that CO_2 concentrations are increasing at an accelerating rate rather than at a constant rate. This acceleration in CO_2 growth means we face a narrowing window for climate action. The findings confirm that current trends are unsustainable and should accentuate the urgency of immediate, aggressive mitigation strategies to prevent crossing critical climate thresholds.